

JAWAHAR R

Aspiring Data Engineer · AI & Data Science Student

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SUMMARY

Data Engineering student who builds and ships production-style pipelines rather than just studying them — including FlowMatrix, a Kafka-to-dbt e-commerce pipeline benchmarked against Rakuten Ichiba's data model, deployed live with automated data-quality checks. Comfortable across ingestion (Kafka, REST APIs), transformation (PySpark, dbt), orchestration (Airflow), validation (Great Expectations), and serving (FastAPI, PostgreSQL, Streamlit). Certified across IBM, Google Cloud, and NPTEL. Targeting a Data Engineering internship where pipeline reliability and clean data modeling matter as much as shipping fast.

SKILLS

- **Languages:** Python, SQL, JavaScript, HTML/CSS
- **Data Engineering:** ETL/ELT, Apache Kafka, PySpark, dbt, Apache Airflow, Great Expectations, Data Modeling, Real-Time Processing
- **Databases & Cloud:** PostgreSQL, Neon Serverless, Docker, Google Cloud Platform, Render.com
- **Frameworks:** FastAPI, Uvicorn, REST API Design, Streamlit **AI/ML:** NLP, Machine Learning, Generative AI

EXPERIENCE

Data Analytics Virtual Internship

AICTE + EduSkills, supported by Google Cloud | Jun – Aug 2026 (8 weeks)

- Completed an 8-week Data Analytics Virtual Internship under the National Internship Portal (AICTE + EduSkills), supported by Google Cloud.

AI-ML Virtual Internship

AICTE + EduSkills, supported by Google for Developers | Oct – Dec 2025 (10 weeks)

- Completed a 10-week AI-ML Virtual Internship; presented the Live City Pulse data pipeline at the national AI For Bharat Hackathon and college project expo.

PROJECTS

FlowMatrix — Real-Time E-Commerce Data Pipeline | Kafka, PySpark, dbt, Airflow, Great Expectations, PostgreSQL | flowmarics.streamlit.app

- Built an end-to-end ELT pipeline (Kafka → PySpark → dbt → PostgreSQL) modeled on the Olist dataset as a Rakuten Ichiba proxy; processed 1,247 streaming events into 7 dbt models with 34 automated data-quality tests (100% pass rate).
- Orchestrated ingestion-to-dashboard refresh via a 6-task Airflow DAG with automated retry logic, achieving end-to-end pipeline runtime under 2 minutes; deployed a live Streamlit front-end backed by PostgreSQL (Neon Serverless).

Real-Time 3D Asset Data Pipeline & Serving API | Python, FastAPI, Three.js, NLP

- Built an ETL pipeline with FastAPI: streamed binary .glb assets over HTTP, parsed prompts via NLP keyword-matching into metadata queries, and served them through a thread-safe cache, cutting repeat-load latency to near-zero.

Weather ETL Dashboard | JavaScript, OpenWeatherMap API • Kisan-Sahayak — AI Agricultural Assistant | Python, NLP, Translation API

- Client-side ETL dashboard streaming live weather/AQI data for 5 cities, zero-backend, deployed on GitHub Pages.
- AI assistant (AI For Bharat Hackathon, national-level) delivering real-time crop prices and farming advice in Tamil and Hindi via an NLP translation pipeline.

CERTIFICATIONS

- **Introduction to Data Engineering** — IBM / Coursera — May 2026
- **Python for Data Science, AI & Development** — IBM / Coursera — Jul 2026
- **Python Project for Data Engineering** — IBM / Coursera — Jul 2026
- **Google Cloud Data Analytics Certificate** — Google Cloud — Jul 2026
- **Computer Architecture & Organization** — NPTEL / IIT Kharagpur — Jul–Oct 2025

EDUCATION

B.Tech, Artificial Intelligence & Data Science | 2024 – 2028 | Dhanalakshmi Srinivasan Engineering College (Autonomous), Perambalur, Tamil Nadu

CGPA: 8.5 (through 4th semester)

LANGUAGES

Tamil (Native) • English (Advanced) • Japanese (Beginner — preparing for JLPT N5)